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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 – 2027

TITLE OF THE PROJECT	**ArchVision AI: Intelligent Architecture Design Assistant**	
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Abstract

ArchVision AI – An Intelligent Floor Plan Analysis and House Design Assistant is an AI-powered architecture support system designed to assist users and architects throughout the residential building design process. Traditional house planning often requires significant manual effort for analyzing floor plans, checking design requirements, estimating construction costs, and selecting suitable materials. ArchVision AI aims to simplify these processes by integrating computer vision, artificial intelligence, rule-based analysis, and full-stack web technologies into a single platform.

The proposed system allows users to upload a house floor plan in image or PDF format. Using computer vision and AI techniques, the system analyzes the uploaded plan to identify and segment rooms, walls, doors, windows, and other architectural components, while extracting available dimensions and spatial information. The extracted information is converted into a structured representation of the house layout, which is then analyzed for space utilization, room dimensions, accessibility, and basic building-code compliance based on predefined rules.

In addition to floor plan analysis, ArchVision AI provides an estimated construction cost based on built-up area, room requirements, and selected materials. The system also evaluates the design using a sustainability score and recommends suitable materials based on factors such as cost, durability, and environmental impact. Based on the analysis, the system can provide design improvement suggestions and alternative layout recommendations. Finally, the platform generates a professional report containing the floor plan analysis, compliance results, estimated cost, sustainability assessment, material recommendations, and design suggestions.

The system can be implemented using **React.js** for the frontend, **Spring Boot** for backend services, **PostgreSQL** for data management, and **Python with OpenCV/PyTorch** for AI-based floor plan analysis. Cloud services can be used for deployment and storage. The proposed system provides a practical, intelligent, and user-friendly approach to residential architectural planning by reducing manual analysis, improving decision-making, and bringing multiple stages of the house design process into a unified platform.

Signature of the Guide HOD